

Townsend: 11.15, 12.2, 12.4, 12.8

Townsend Problem 11.15 (5 pts)

Determine the effect of an external magnetic field on the energy levels of the $n = 2$ states of hydrogen when the applied magnetic field B has a magnitude much greater than 10^4 gauss, in which case the spin-orbit interaction may be neglected as a first approximation. This is the **Pashen-Bach effect**.

Draw an energy-splitting diagram like Figure 11.10.

Hint: Unlike the section problem and most of the section on the Zeeman Effect, you can start with the (11.105) form rather than the (11.106) form of the perturbing Hamiltonian.

■

Townsend Problem 12.2 (5 pts)

Two identical, noninteracting spin- $\frac{1}{2}$ particles of mass m are in the one-dimensional harmonic oscillator for which the Hamiltonian is

$$\hat{H} = \frac{\hat{p}_{1x}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}_1^2 + \frac{\hat{p}_{2x}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}_2^2$$

- (a) Determine the ground-state and first-excited-state kets and corresponding energies when the two particles are in a total-spin-0 state. What are the lowest energy states and corresponding kets for the particles if they are in a total-spin-1 state?
- (b) Suppose the two particles interact with a potential energy of interaction

$$V(|x_1 - x_2|) = \begin{cases} -V_0 & |x_1 - x_2| < a \\ 0 & \text{elsewhere} \end{cases}$$

Argue what the effect will be on the energies that you determined in (a), that is, whether the energy of each state moves up, moves down, or remains unchanged. *Suggestion:* Examine which spatial wave functions for the total-spin-0 and total-spin-1 states tend to have the particles closer together. Consider, for example, the special case of $x_1 = x_2$.

■

Townsend Problem 12.4 (5 pts)

Use the variational principle to estimate the ground-state energy for the one-dimensional anharmonic oscillator

$$\hat{H} = \frac{\hat{p}_x^2}{2m} + b\hat{x}^4$$

Compare your result with the exact result

$$E_0 = 1.060b^{1/3} \left(\frac{\hbar^2}{2m} \right)^{2/3}$$

■

Townsend Problem 12.8 (5 pts)

Use the trial wave function $R(r) = Ne^{-br}$ to estimate the ground-state energy of the three-dimensional isotropic harmonic oscillator, for which $V(r) = \frac{1}{2}\mu\omega^2 r^2$. Compare your estimate with the exact value (see Section 10.5).

■